

The Iterated Prisoner's Dilemma

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1 Introduction

The Prisoner's Dilemma is one of the most well-known problems in game theory. It describes a situation in which players must independently choose between cooperation and defection. Although cooperation can lead to a better outcome for both players, each player has an individual incentive to defect. This creates a conflict between individual incentives and the outcome that is collectively beneficial.

The Prisoner's Dilemma has been studied in many different forms, particularly when the game is played repeatedly. This paper will examine how different conditions of the game affect the outcomes observed in existing research. In particular, it will consider how factors such as the number of rounds, payoff structure, number of players, and strategies being used can influence the behaviour and performance observed in the game.

2 Background

2.1 The Prisoner's Dilemma

The Prisoner's Dilemma is a game in which players independently choose between two possible actions: cooperation and defection. The payoff received by each player depends on both their own action and the action chosen by the other player.

For a two-player Prisoner's Dilemma, the payoffs can be represented by the following matrix:

	C	D
C	(R, R)	(S, T)
D	(T, S)	(P, P)

where C represents cooperation and D represents defection. The four payoff values are usually required to satisfy

$$T > R > P > S.$$

Here, T is the temptation payoff received by a player who defects while the other player cooperates, R is the reward received when both players cooperate, P is the punishment received when both players defect, and S is the sucker's payoff received by a player who cooperates while the other player defects.

One commonly used set of payoffs is

$$T = 5, \quad R = 3, \quad P = 1, \quad S = 0.$$

Under these conditions, defection is a dominant strategy. Regardless of the action chosen by the other player, a player receives a higher payoff by defecting.

Therefore, when the game is played only once, both players choosing to defect forms a Nash equilibrium. However, mutual cooperation would give both players a higher payoff than mutual defection. This conflict between individual incentives and the collectively better outcome is the central dilemma of the game.

2.2 The Iterated Prisoner's Dilemma

The Prisoner's Dilemma can also be played repeatedly between the same players. This is known as the iterated Prisoner's Dilemma.

Instead of making a single decision, players make decisions over a sequence of rounds. The outcome of one round can therefore provide information that can be used when making a decision in a later round. This introduces the possibility that a player's current behaviour can depend on the previous behaviour of their opponent.

For example, a player may cooperate in an initial round and then continue cooperating if their opponent also cooperates. Alternatively, they may respond to a previous defection by defecting in the next round.

The repeated nature of the game therefore introduces strategies that are not relevant in the one-shot version. A strategy in the iterated Prisoner's Dilemma can be viewed as a rule that determines a player's action based on the information available from previous rounds.

2.3 Strategies in the Iterated Prisoner's Dilemma

Different strategies can produce very different behaviour when repeatedly interacting with one another.

Some simple examples are:

- **Always Cooperate:** the player cooperates in every round.
- **Always Defect:** the player defects in every round.
- **Tit-for-Tat:** the player cooperates in the first round and then copies the opponent's action from the previous round.

These strategies illustrate why the iterated version of the Prisoner's Dilemma is different from the one-shot version. For example, two players using Always Defect will defect in every round, while two players using Tit-for-Tat will cooperate in every round if they both begin by cooperating.

When different strategies interact, however, the sequence of outcomes can become more complicated. A strategy that performs well against one opponent may not perform equally well against another.

2.4 Why the Iterated Prisoner's Dilemma Is Interesting

The iterated Prisoner's Dilemma provides a way to study how behaviour changes when players interact repeatedly rather than only once.

One important area of interest is cooperation. Although defection is the dominant strategy in the one-shot game, repeated interaction can allow cooperation to persist through strategies that respond to previous behaviour.

Another important aspect is the interaction between strategies. The outcome of a game does not depend only on the strategy used by one player. It can also depend on the strategy used by the opponent. Consequently, the performance of a particular strategy may change when it is placed in a different strategic environment.

These observations raise questions about whether conclusions about the Prisoner's Dilemma are consistent across different ways of setting up the game.

2.5 Conditions Studied in Previous Research

Researchers have studied the Prisoner's Dilemma under a wide range of conditions. Rather than treating the game as having one fixed setup, different studies have changed various aspects of the game to investigate how these changes affect its outcomes.

Some of the conditions that have been varied include:

- **Number of rounds:** outcomes may change based on the number of rounds that are played.
- **Payoff structure:** the values of T , R , P , and S can be changed, altering the incentives associated with cooperation and defection.
- **Number of players:** although the traditional Prisoner's Dilemma is a two-player game, researchers have also considered versions involving more players.
- **Strategies:** studies can compare different sets of strategies, which can affect the relative performance of the strategies being investigated.
- **Noise and errors:** some studies allow players to make errors, meaning that the action actually played may differ from the action intended by the player.
- **Ending conditions:** repeated games can differ in whether the number of rounds is known in advance or whether there is uncertainty about when the game will end.

The following sections examine research involving some of these different conditions and compare the effects that they have on the outcomes of the Prisoner's Dilemma.

3 Effect of Game Conditions

3.1 Number of Rounds

The number of rounds can affect the way players behave in the Iterated Prisoner’s Dilemma because repeated interaction gives players more opportunities to respond to their opponent’s previous actions. This also means that players may behave differently depending on how far they are into the game. Two studies that help illustrate this are Wang et al. (2016) and Mao et al. (2017).

Wang et al. (2016) directly investigated the effect of game length by having participants play games consisting of either 60 or 500 rounds. Participants played against computer-controlled strategies, including an *extortionate* strategy and a *generous* strategy. The researchers also varied whether participants knew that their opponent was a computer.

One of their results was that cooperation increased over the course of the 500-round games when participants knew they were playing against a computer. Cooperation can be measured for each participant as the proportion of their decisions that were cooperative. For each participant, the researchers compared this cooperation rate during the first 100 rounds with their cooperation rate during the final 100 rounds. The average cooperation rate increased from approximately 0.563 in the first 100 rounds to approximately 0.757 in the final 100 rounds. A Wilcoxon signed-rank test found this increase to be statistically significant ($n = 32$, $z = 4.207$, $p < 0.001$).

The Wilcoxon signed-rank test is appropriate here because the two cooperation rates come from the same participants: each participant provides a pair of observations, one from the beginning and one from the end of the game. The test examines whether the differences between these paired observations are consistently different from zero, without requiring the cooperation rates to follow a normal distribution. Thus, the statistical result supports the conclusion that cooperation increased over the course of these 500-round games.

The researchers also found that the extortionate strategy outperformed the generous strategy in the 500-round condition when participants knew they were playing against a computer, whereas this difference was not observed in the 60-round condition. This comparison used a Mann–Whitney test to compare outcomes between the different treatment groups.

These results suggest that longer interactions can change the development of cooperation. However, they do not mean that simply increasing the number of rounds always produces more cooperation. The 500-round games gave participants much more time to learn and adapt to their opponent, and the experiment also varied whether participants knew the opponent was a computer. Therefore, the effect of game length cannot be completely separated from these other factors.

Mao et al. (2017) examined a different aspect of repeated interaction. Instead of comparing different game lengths, they kept each game at 10 rounds and looked at how behaviour changed within a game and across many repeated games. Their experiment involved 94 participants who played up to 400 ten-round games over 20 consecutive weekdays, with participants being randomly assigned new partners after each game.

The researchers focused on the *round of first defection*. For each game, this is the first round in which a player defects; if neither player defects, the game is treated separately.

This gives a random variable that describes where defection first occurs within the ten-round game. They examined the distribution of this variable on each day of the experiment.

During the first several days, the distribution of the round of first defection shifted towards earlier rounds. In other words, players who initially waited until later rounds to defect increasingly began defecting earlier. The researchers used a two-sample Kolmogorov–Smirnov (K–S) test to compare the distributions of first-defection rounds on successive days. The changes were statistically significant through approximately day 7, after which the changes were no longer statistically significant. They therefore identified approximately day 7 as the point at which this “unravelling” process had stabilised.

Together, these studies show that the number and position of rounds can matter, but its effect is not simply a matter of “more rounds means more cooperation” or “more rounds means less cooperation.” Wang et al. show that behaviour can change substantially over the course of a long game, while Mao et al. show that repeated experience can cause behaviour within a finite game to change over time before eventually stabilising.

3.2 Payoff Structure

The payoff structure of the Prisoner’s Dilemma determines how attractive cooperation and defection are to the players. The four payoffs are usually denoted by T , R , P , and S , where T is the payoff for defecting when the other player cooperates, R is the payoff when both players cooperate, P is the payoff when both players defect, and S is the payoff for cooperating when the other player defects. For the game to be a Prisoner’s Dilemma, these payoffs satisfy

$$T > R > P > S.$$

Therefore, a player always has a higher individual payoff from defecting than cooperating, regardless of what the other player does. However, mutual cooperation gives both players more than mutual defection because $R > P$.

Gächter et al. (2024) investigated whether changing these payoff values affects people’s willingness to cooperate. They conducted three experiments involving a total of 1,993 participants and varied the payoffs across different one-shot Prisoner’s Dilemma games. Their main experiments were designed so that different aspects of the payoff structure could be varied separately, allowing them to investigate which features of the payoffs were related to cooperation.

One of the main quantities they considered was the efficiency gain from cooperation, which measures how much better mutual cooperation is compared with mutual defection. This is captured by the normalized payoff difference

$$\text{EFF} = \frac{R - P}{R}.$$

They also considered the temptation to defect and the risk of cooperating when the other player defects.

The researchers found a robust relationship between the efficiency gain and cooperation. When the gain from mutual cooperation over mutual defection was larger, participants

were more likely to cooperate. They analysed cooperation as a binary outcome, where each participant either cooperated or defected, and used regression models to examine how the probability of cooperation changed with the payoff indices. In this analysis, the regression estimates how changes in a payoff feature such as EFF are associated with changes in the probability of cooperation, while allowing the different payoff features to be considered separately. An increase in EFF of 0.1 was associated with a 3.99 percentage-point increase in the probability of cooperating in one of their analyses. In contrast, the effect of the risk of being exploited was not statistically significant, while the effect of the temptation to defect was less consistent across their experiments.

This suggests that changing the payoff structure does not affect cooperation simply because one particular payoff becomes larger. Instead, what matters is the relationship between the different payoffs and, in particular, how attractive mutual cooperation is compared with mutual defection.

3.3 Strategies

The strategy used by a player determines how they respond to their opponent's actions throughout the Iterated Prisoner's Dilemma. Different strategies can therefore produce very different levels of cooperation even when the payoff structure and number of rounds are the same. One well-known strategy is *Tit-for-Tat*, which cooperates on the first move and then copies the opponent's previous action.

Axelrod (1980) investigated how different strategies perform against one another through a computer tournament. In the first tournament, strategies submitted by experts in game theory and related fields played against one another in a round-robin tournament. Each strategy was evaluated according to the score it accumulated against the other strategies. Tit-for-Tat achieved the highest overall score.

The performance of a strategy was measured using its score across the different games it played. If S_{ij} denotes the total score obtained by strategy i when playing against strategy j , then the average score of strategy i across its opponents can be written as

$$\bar{S}_i = \frac{1}{N} \sum_{j=1}^N S_{ij},$$

where N is the number of opponents. Strategies could therefore be compared using their average scores, with a higher value indicating better overall performance in the tournament.

Axelrod then repeated the tournament in a second study, allowing participants to design their strategies after seeing the results of the first tournament. The second tournament contained 62 strategies, providing a test of whether successful strategies from the first tournament would remain effective after participants had learned from the earlier results. Tit-for-Tat again achieved the highest overall score.

To investigate why some strategies performed better than others, Axelrod compared the characteristics of the strategies with their tournament performance. The analysis showed that successful strategies tended to have three properties. They were *nice*, meaning that they did not defect first; *provocable*, meaning that they responded to defection; and *forgiving*,

meaning that they returned to cooperation when the opponent cooperated again. Tit-for-Tat has all three properties.

These results suggest that the strategy used by a player can strongly affect cooperation and accumulated payoff in the Iterated Prisoner's Dilemma. They also show that a strategy's performance depends on how it interacts with the strategies used by other players. However, Axelrod's tournaments do not establish that Tit-for-Tat is always the best strategy, since its performance was measured under the particular payoff structure, tournament design, and set of competing strategies used in these experiments.

4 Research Directions

The Prisoner's Dilemma can be extended beyond the conditions considered in this paper. For example, the game could involve n players instead of two, or players could have more than two possible actions. Errors could also be introduced by allowing a player's intended action to be incorrectly implemented with probability p . These extensions could be studied individually or together to investigate how cooperation changes as the game becomes more complex. The standard Prisoner's Dilemma corresponds to the special case of two players, two actions, and no errors.

References

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